

Slowdown Theorem on Links



Work in Progress – Paul Borrill [Daedaelus](#)*

1 Introduction

This document is in response to a conversation between Edward Lee, Jonathan Gorard and Paul Borrill on 02023-May-12. And, a request by Jonathan to describe our link protocol in the simplest possible mathematical terms. We do this in the context of Jonathan's Slowdown Theorem [1], and using the language of Beggs and Tucker [2] referenced therein.

A discrepancy in perspective may be due to hidden assumptions:

- (1) One-sided (Forward In Time Only) perspective for SME experiments.
- (2) Not explicitly including reversibility as the counterfactual,
- (3) Unconsciously relating to a smooth background of spacetime, and
- (4) An implicit assumption of Cellular Automata (CA) synchronicity.

We deal with each of these assumptions, and offer an alternative model which could potentially explain indefinite causality, and be useful in implementations of Lingua Franca in CAL registers in a SmartNIC.

We all know that the big challenge is understanding what Lucien Hardy, Časlav Brukner and Giulia Rubino have described in the Its About Time club as the "wild beast of indefinite causality".

My primary concerns in Jonathan's Wedge Scattering experiment, and Edward's description of CAL registers, is they do not account for either "indefinite causality" or "reversibility" as critical to our understanding of time. As Lucien Hardy says, we are 'missing half the picture' [3].

This appears to be similar to Stephen Wolfram's models of CA on a single machine. And, the same issue that the Stanford folks have in trying to synchronize clocks (see lecture notes on Stanford 16-May-2023 Lecture in iCloud Folder). We call this the "Minkowski Error." Which we describe briefly here, and in more depth in a separate document.

2 Slowdown Theorem

In [1] Jonathan proves – The Slowdown Theorem: A Lower Bound for Computational Irreducibility in Physical Systems.

Tucker and Gorard proved a theorem about scattering machine experiments where they showed that in standard Newtonian physics, imagine a very simple experiment where you have a projectile that's fired at a wedge, and the wedge reduces to an infinitesimal point. And it can either scatter up or it can scatter down. You've reduced the setup of this experiment to just one of two possible binary observables: upscatter or downscatter. They showed that by tuning the system in an appropriate way, with only a finite amount of computational effort that you could effectively excavate, arbitrarily many decimal places in the initial conditions. And you could use this in principle to do hyper computation.

In his analysis, Jonathan does not go from:

$$f : \mathbb{R} \rightarrow \mathbb{N}.$$

He assumes that systems are discrete, which implies digital physics:

$$f : \mathbb{N} \rightarrow \mathbb{N}.$$

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[3]: Hardy (2021), *Time Symmetry in Operational Theories*

The Slowdown Theorem: A Lower Bound for Computational Irreducibility in Physical Systems

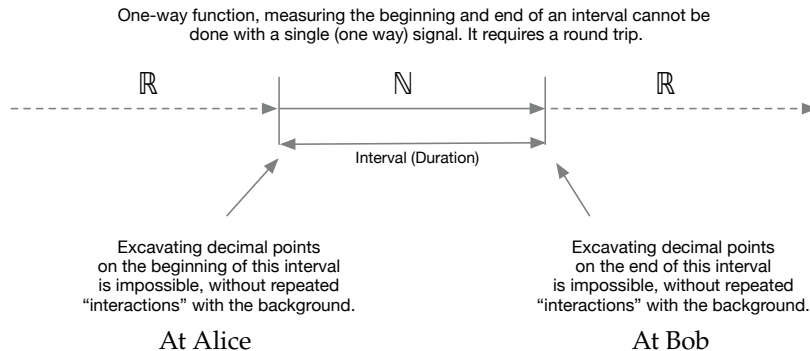
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Following from the work of Beggs and Tucker on the computational complexity of physical oracles, a simple diagonalization argument is presented to show that generic physical systems, consisting of a Turing machine and a deterministic physical oracle, permit computational irreducibility. To illustrate this general result, a specific analysis is provided for such a system (namely a scatter machine experiment (SME) in which a classical particle is scattered by a sharp wedge) and proves that it must be computationally irreducible. Finally, some philosophical implications of these results are discussed; in particular, it is shown that the slowdown theorem implies the existence of classical physics experiments with undecidable observables, as well as the existence of a definite lower bound for the computational irreducibility of the laws of physics. Therefore, it is argued that the hypothesis that "the universe is a computer simulation" has no predictive (i.e., only retrodictive) power.

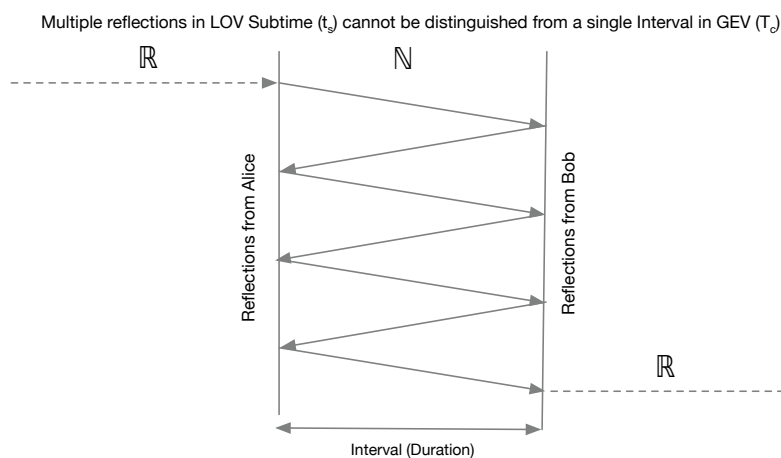
Gorard [1] Beggs & Tucker [2]

3 Event Arrival Discretization

A Perspective on Discretization of Arrival Events:



Intervals (durations) cannot be measured on a real line. (Forward In Time Only (FITO) Thinking.



Intervals (durations) cannot be measured on a real line. Reflections in Time-symmetric Alternating Causality (photon links) are not subadditive, and not represented by Max-Plus Algebra.

Excavating decimal points on the beginning of an interval requires multiple interactions. At least one round-trip.

Excavating decimal points on the end of an interval requires multiple interactions. At least one round-trip.

Trying to measure (M) in arbitrary Minkowski Spacetime by sending out a photon (projectile) will therefore will be nothing more than 'correlations' without signalling.

One round trip will require a classical measurement of the speed of light (always requires roundtrips to do signaling).

Two round trips will enable "information transfer" – an information token on one side can be reversibly transferred to the other side.

[INSERT FIGURE - Two Wedges Back to Back - use Jonathan's picture]

The classical assumption that the infinitesimal point goes to zero is questionable. Quantum theory would say that the tip has some minimum size, although that's somewhat complicated. We know how to do X-Ray Crystallography, and our electron microscopes show the structure of individual atoms in certain materials.

In reality, there will be cases where the projectile hits *exactly* on this point, and reflects *exactly* down the reciprocal path. Now if this happens, the ping pong goes on 'in-perpetuity'. This is sub-causal behavior. Lost when you do coarse graining. You have lost the particle to *subtime* [4].

The big question for experimental science is are these models observationally isomorphic?

In Jonathan's analysis $T^* \equiv$ computes some value of $f(i)$ for some fixed input i in n steps. But this assumes that each step is monotonic, and no 'stuttering' in subtime.x

4 Beggs & Tucker

The link is our version of the "the equipment" in the language of Beggs & Tucker[2].

"the actions allow construction, alternation, initialization, and observation of the equipment."

"The procedure schedules, the actions, step-by-step" not in time, measured by clocks, but in events, occurring internally in the link that cannot be measured (observed) outside of the link, without stealing the mechanism that is doing the ticking and tokking within "the equipment".

page 9: describes 3 outcomes: r , l , o . What they don't seem to appreciate is that r (right collecting box), and l (left collecting box) create events which the detector can use to trigger a measurement. The third outcome " o " does not. The projectile has gone off into never-never land and is never seen again.

However, there is a 4th case, not considered by Beggs & Tucker: that the projectile gets trapped within the equipment, bouncing back and forth like a ping pong ball between the 'points of two wedges'. This 4th potential outcome initially appears indistinguishable from the 3rd¹.

In both outcome 3, and outcome 4, the detectors are 'dark'. #3 because the particle escaped, and #4 because it was 'trapped', inside the equipment.

We can, however, detect the difference between outcomes #3 and #4: if you send another projectile through the apparatus, the 'trapped' projectile will be displaced and come out the other side.

i.e. The trapped projectile case is a 'resource' that is hidden inside the equipment, which can be used to guarantee 'exactly-once' operation. The mathematics of which will be identical to the no-cloning, and no broadcasting theorems in quantum mechanics.²

o must generate an event

Beggs & Tucker rely on an external clock to discover missing events, just like Edward does in *Lingua Franca*. We will prove (in a separate analysis) why this won't work for *Lingua Franca*, unless they have the kind of mechanisms we are proposing in the SmartNICs to 'harvest' 'alternating causality' events from other³ sources, e.g. the cell agent sitting on top of all 8 links, when a packet gets lost.⁴

5 Hidden Reversibility

The big problem with reversibility is when does it stop? (Loschmidt's paradox).⁵ There are several ways to think about it:

1. It all goes in the same direction until the end of the universe, and at that point it reverses, and traces a path back through the universe from the big crunch to the big bang.
2. It is completely time symmetric in the Schrödinger equation.
3. It is completely time symmetric in General Relativity, so should we wait for the end of the universe for the Janus Point [6]
4. Is there an alternative explanation that does away with this human imagined arrow of time that is consistent with our measurements in the laboratory, and with quantum information Theory?

[2]: Beggs et al. (2007), *Experimental computation of real numbers by Newtonian machines*

Computation Theory is dominated by "Forward In Time Only" (FITO) concepts such as cannons and collecting boxes

1: This is one reason why constructing an experiment and how difficult it is to guarantee 'exactly one' photon emission, and 'exactly one' photon detection. This is discussed in the subtime paper [5].

2: By the way, this notion of sending cannonballs through the apparatus can be used by Pat Helland to create flexible quorums, which is the way I suspect he tries to solve the problem in a public cloud like AWS.

3: Absence of Evidence is not Evidence of Absence

4: See FAQ-Link, and other documents in the DAE FAQ set for a description of what we do with runt and other corrupted packets – we don't discard them like conventional networks do – we harvest them for the events they create, which allows us to invoke recovery mechanisms ... such as sending the state machine backwards in time, to a consistent state where information is never lost

5: In a photon clock the reversibility goes on forever but an external 'perturbation' dislodges the photon from the entangled pair and it carries on it's merry way.

[6]: Barbour (2020), *The Janus point: A new theory of time*

Should we treat "information surprisal" as the measure of time going forwards (we receive information, but then that information is immediately obsolete). It seems obvious to me that information is a conserved quantity. If the receiver 'increases' their information, then the sender 'decreases' their information.

This is consistent with Spekkens' Knowledge Balance Principle (KBP)

The big thing we learned from the ItsAboutTime.club, is that relativity (SR and GR) alone are insufficient to describe spacetime. Without some modeling of indefinite causality, we will never be able to create a correct model of spacetime and achieve "determinism on demand" in our distributed systems software.

Our model follows the rewriting systems (Abstract Rewriting Rules) of Wolfram and others, but with a twist: there exists a second mechanism where the causality is reversed at each reflection, and the rewriting rules occur in both the forward Alice $\rightarrow$ Bob, and Bob $\rightarrow$ Alice event structure.

Partial Function

Initial conditions (copied from B & T)

- (i) "Input data x are used to determine initial conditions of the physical system (Link)
- (ii) The system operates for a finite or indefinite time;
- (iii) Output data y are obtained "by measuring the observable behavior of the system"

TO WHICH WE ADD:

Partial Function

- (i) The above partial function is incomplete. We need two partial functions: one where x is sent through the system, and produces y , and one where y is sent it through the system and produces x .
- (ii) Those two partial functions are tied together, such that y transformed causes y' to be sent on the wire, and x transformed causes x' to be sent on the wire.
- (iii) connecting y' (sent) by a wire to x (receiver), and x' (send) to y creates a continuously circulating (Alternating Causality) link.

Note that, unlike B & T, the system cannot be observed without breaking the circuit. The system cannot simply "read". Because "read" is destructive. The Petri token is conserved, and either continues to be circulated, or it is redirected to cause an event outside (kicked upstairs).

But how do we get any information "out" from say side y ?

The answer, is you put the information in on side $x \rightarrow$ prior to doing the transform from x to x' , the Petri circuit tests (using ancilla to define a change in rewriting rule) to see if there are tokens in a different bucket $\rightarrow$ then I need to send a different operation (colored token)

The same is true. If you want to get information "out" of side x , you have to put in information on side y . This "conservation principle" is newtons cable⁶.

6: A whimsical take on Newtons cradle.

6 Asymmetric Scatter Machine

Beggs & Tucker describe an Experimental computation of real numbers by Newtonian machines [2] – we refer to this as Asymmetric Scatter Machine (ASM), because the initial conditions (initial orientation in space – and time) is prescribed before the experiment begins.

[2]: Beggs et al. (2007), *Experimental computation of real numbers by Newtonian machines*

Interaction

We need a robust notion of interaction, so if $f : \mathbb{R} \rightarrow \mathbb{R}$ are reciprocal operations in a closed causal loop, this means:

$\mathbb{N} \rightarrow \mathbb{R} \rightarrow \mathbb{N}$ is the one-way circuit from Alice to Bob, and

$\mathbb{N} \leftarrow \mathbb{R} \leftarrow \mathbb{N}$ is the one-way circuit from Bob to Alice (from Alice's perspective), then what Daedaelus does in its link protocol is create a circulating loop from these two components, with a rewriting function (transformation) $\mathbb{N} \rightarrow \mathbb{N}'$ on both ends of the link.

To avoid this problem, Jonathan assumes digital physics: $f : \mathbb{N} \rightarrow \mathbb{N}$

This is also a one-way function (FITO).

The protocol space is small. The rules are simple, but as with all simple rules, they can quickly create complex behavior. This is why we need Jonathan, and Stephen, to design the hierarchical toolset to allow us to systematically explore this space in Mathematica.

Senders and Receivers

$\mathbb{N}_{AS} \rightarrow \mathbb{R} \rightarrow \mathbb{N}_{BR}$

Alice Sender is connected to Bob's Receiver through an asynchronous interface.

$\mathbb{N}_{BS} \rightarrow \mathbb{R} \rightarrow \mathbb{N}_{AR}$

Bob's Sender is connected to Alice's Receiver, through another asynchronous interface.

But, from a GEV perspective:

$\mathbb{N}_A \leftrightarrow \mathbb{N} \leftrightarrow \mathbb{N}_B$

Is timeless, the direction of time (and causality) is local (LOV, hidden from the GEV).⁷

The only way to assert causality through this link is by kicking it from one side or the other⁸.

7: Liouville's Theorem "The density of points representing particles in a 6D phase space is conserved as one follows them through that space".

8: The AC protocol is hidden, silent, very much like the picture of Newton's Cradle, where the center balls remain still and only the outer ones oscillate – like a metronome.

7 Scatter Machine Experiment

Note: The language of Inputs and Outputs need to be replaced with Hardy's incomes and outcomes, so they are fully reversible from a computational basis [3].

We know that information is conserved. Why do we not talk about the + and – relationship in information between the receiver and the sender? (See Subtime document).

The point cannot be infinitely sharp. This is the metastability logic error that Ivan Sutherland made when I was at Sun Microsystems.

Let's use Jonathan's Language.

- Decoding Function
- Encoding Function

Note that the above concepts are FITO thinking⁹. This means that you can take the Decoding function, and run it backwards to create an encoding function. And you can take the encoding function and run it backwards, to create a decoding function.

This is the 'symmetric' nature of the Alice and Bob (or Alice and Alice') perspectives.

[3]: Hardy (2021), *Time Symmetry in Operational Theories*

9: FITO = Forward In Time Only thinking – i.e. missing the counterfactual.

You can't do this to your self (i.e. in a single computer), it has to be a bipartite relationship with some notion of mathematical reflection. Perhaps a conjugate does this, plus some notion of adjoint? As Jonathan suggested on Friday?

"Any discussion of initial and final states is non-computable"
(From B & T).

From Jonathan: "After colliding with the wedge, the particle with either scatter upward (into detector 1), or downward (into detector 2)"

"For the sake of simplicity, we, we shall henceforth assume that the experiment is configured in such a way that the particle never hits the point on the wedge.¹⁰"

A single binary variable.

10: That's a little bit too convenient

"In which point collisions are not permitted"

POINT TO POINT COLLISIONS ARE THE NORM, NOT THE EXCEPTION.

In a world where (time is change you can count), time (complexity) does not accumulate unless it can overcome the endless entropy of photons scattering off everything forever.

When you permit point collisions, you enter Subtime [4] and all relationships with the outside world ceases. From the outside, time stops.

[4]: Borriell (2013), 'Time One: The Subtime Conjecture'

Yes, you can hit a point, and when you do, that's all you can do, keep hitting the point in the symmetric scatter machine.

P 182: For alternating causality there are one step at a time on both sides.

Correctness (C) is not a one shot set of n steps.

You cannot make a particle move at arbitrarily low speed. All photons travel at exactly the speed of light¹¹.

You can't gain any benefit from moving the wedge further away from the cannon: With Quantum time, you get phase only, which carries with it, no information on distance travelled.

11: It is easy to show that photons traveling through a material have a 'slow-down' according to the Causal Square (Discussed Later), resulting in an experimental observation of refractive index.

P 182 figure. COPY THIS AND DO BACKWARDS

"The SME is just a particularly elegant minimal example of such an experiment E for which computational irreducibility holds."

BUT YOU ARE MISSING THE COLLISIONS BETWEEN THE POINTS!

Interactions Again

Using the terminology from Pat Helland (Autonomous), we define:

- ▶ An *Interaction* is bidirectional sequence of messages obeying the knowledge balance principle.
- ▶ "Each interaction in a collaboration starts by digging deeper into the solution, and completes by digging back out".¹²

There is no notion of time. It only increases and decreases in information loss and information gains.

12: This is what we mean by symmetry of 'information' in an *interaction*.

For Rosetta Stone paper: Imagine a Netronome, put it in a soundproof box. There are two Maxwell's twins, but they are outside the door at each end of the box, when one of the opens the door, the sound pressure wave comes out that end. But we are out on the surface of the moon and the air comes out, and the demon has to run to the other end of the box in order to tell his twin what he saw.

This concept of slowdown is non-negative. It hides a FITO assumption underneath. This is related to the unmeasurability of "durations".

From Jonathan & Edward Conversation

The universe is Turing compatible has no predictive power"

Back to B & T Description

The scatter machine is based on a subtheory T of Newtonian Mechanics, comprising the following:

1. Point particles obeying Newtons Laws of motion in the two dimensional plane.
2. Straight line barriers, with perfectly elastic reflection of particles.
3. Cannons, which can be moved in position, and can project a particle with a given velocity in a given direction.
4. Particle detectors, capable of telling if a given particle has crossed a given region of the plane.
5. A clock to measure time.

(1) The hidden assumptions in the setup is in it's pre-determined asymmetry.

(2) The background of time implied by the use of an external clock.

In nature, Cannons and points are the same thing. Projectiles (photons) can be fired from points, and collected by Cannons.

Analysis of B & T

This is an interesting paper, and the whole reasoning represents "forward in time only" (FITO) thinking: with cannons firing projectiles against a wedge shaped object, and cannon balls entering the left collecting box, or the right collecting box. The intellectual exercise, is what happens at the point?

They explicitly state that the behavior is deterministic on either side of the wedge, but is *not defined* on the exact point of the wedge¹³.

B&T conclude that "non-computability is to be found in its initial state and output".

They evaluate all permutations of the ball landing in the left and right collecting box, and the 'o' no ball in either.

They correctly identify the issue, but don't resolve it in the Asymmetric Scatter Machine (ASM) Experiment setup.

Jonathan also considers the same reasoning, and does not consider the case where the particle hits the point. ("Point collisions are not permitted").

13: This is exactly the same as the error made by Ivan Sutherland in thinking about asynchronous systems, and the resolution to metastability on a real line.

8 Symmetric Scatter Machine

SSM: We now consider a similar set-up to Beggs & Tucker, with the following:

1. Photons¹⁴. obeying Wheeler & Feynman's Electrodynamics in two dimensional plane.
2. Straight line mirrors, with perfect reflection of particles, at 45 degrees.
3. Symmetric sources emission and absorption (hydrogen atoms).
4. Photon detectors, capable of telling if a given photon has arrived..
5. No clocks: Photons are the carrier of time. [5]

14: projectiles that only travel at c

[5]: Borrell (2013), *An Insight Into Information, Entanglement and Time*

9 Minkowski Error

I propose "Minkowski Error" (ME) to describe a perspective of assuming $f : \mathbb{R} \rightarrow \mathbb{N}$ and $f : \mathbb{N} \rightarrow \mathbb{R}$, are embedded in $\mathbb{M}$.

(i.e. assuming a Forwards In Time Only (FITO) measurement, ignoring time reversal. $f : \mathbb{N} \rightarrow \mathbb{R}$ OR $f : \mathbb{R} \leftarrow \mathbb{N}$ depending on your symbolic preference for describing reversibility).

ME exposes a hidden assumption that underlie many concepts in computer science where we inadvertently assume a simultaneity plane "out there" in Minkowski Spacetime. This intellectual vantage point is shared by Leslie Lamport, and everyone who has tried to create some kind of logical timestamps, including Edward Lee and his students¹⁵. It is also the same problem as the Stanford (Balaji) TickTock Networks method of slowing everything down by 80 microseconds, to align with an imaginary simultaneity surface in the future [huygens].

15: We will address Linga Franca in a separate review

huygens

A principle of Alternating Causality (AC) respects that every detector is in constant touch with reality if $f : \mathbb{R} \leftrightarrow \mathbb{N}$. And that discretizing the continuum will create anomalous results, unless spacetime were smooth, irreversible, and monotonic – which is a disprovable hypothesis.

Instead, we know from indefinite causality (Lucien Hardy and Caslav Brukner IAT Sessions, and references), that there is no relationship between $f : \mathbb{R} \leftrightarrow \mathbb{N}$. (The wild beast of indefinite causality).

Instead, the DAE link does the following:

$$i : \mathbb{N}_{a'} \rightarrow \mathbb{R} \rightarrow \mathbb{N}_b \Rightarrow \mathbb{N}_{b'} \rightarrow \mathbb{R} \rightarrow \mathbb{N}_a \Rightarrow \mathbb{N}_{a'}$$

We use i instead of f to indicate an interaction, instead of a one-way function.

This loop from Alice's transmitter, through ONT, followed by reception at Bob's receiver, is then remembered as "the last thing heard from Alice", and transformed $\Rightarrow$ (encoding function¹⁶) into b' before being put back on the wire by Bob's transmitter, going back through ONT, and then into Alice's receiver.

16: Abstract Rewriting Theory

10 One Turing Machine or Two

Jonathan describes a Turing machine T . Turing self steps are not the same as interactions. There may be n steps in a standalone Turing machine, but there are m (mutual) steps in an entangled pair of Turing machines.

In a conventional model for the link, it is assumed that each Turing machine makes some number of steps before sending a message on

the wire to the other Turing machine, and that Turing machine makes some (perhaps different number) of steps before returning a message in response on the wire back to the first Turing machine. We call this RPC.

In our DAE Link we (ideally) take the minimum number of steps on each side to be 1. The definition of this step, is that it *must* be atomic. We will discuss this requirement in detail in our FAQ-Atomicity.

Experimental computation of real numbers on Links

The Link is a computational object, but is it a Turing machine capable of universal computation?

There is no background of time in either Alice or Bob, but they can have a ticking clock between them over an Ethernet link. This 'mutual' clock resembles a photon interaction using constantly circulating packets, containing a pair of encoding and decoding functions for Alice (for the transmit and receive buffers accordingly) and another pair of encoding and decoding functions for Bob.

From one (LOV) perspective,, the encoding and decoding functions are complementary. And can be symmetric. This is reflected in the Alice and Bob "view" of the Petri net link in the DSPS notebook[8].

From another (GEV) perspective, they cannot be symmetric, unless there is a dual (pair) of tokens, then they can recover their symmetry, but still retain the ability to control their own thread of execution (serializability).

- ▶ Each 'token' wins the race to get their 'initialize' packet to the other side of the link first. This is the only time that an internal timer (clocked counter) needs to be used, because without the return of the packet from the other side, the link would stall. The timer can be set to any application-dependent value.¹⁷.
- ▶ When the initial symmetry is broken by which cell wins the first initialization process, this packet remains "owned" by the source (e.g. Alice) and she refers to her partner as Alice'.
- ▶ Maximum redundancy (and flexibility for recovery actions) occurs when there are two packets in flight, one owned by Bob, and one owned by Alice. See The DSPS, State Machine, and Slice Engine specifications.

[8]: Borrill (2022), *Quantum Ethernet*

17: For fast real-time recovery, it might be a microsecond. For attoprobewarms on their way to Proxima Centauri, where energy is at a premium, it might wake up only every 24 hours or so to align with transmission and reception on Earth.

Barbour and Confined Boxes

In [9] Julian Barbour explains the problem of physics in a box in his critique of Rovelli's views.

What is impossible to explain by the known time-reversal symmetric laws of nature is the origin in confined systems of the special initial state.¹⁸ In the theory of dynamical systems, those 'in a box' correspond formally to ones that have a phase space of bounded Liouville measure; to my knowledge systems with phase spaces of unbounded measure have never been seriously studied in statistical mechanics. This is a remarkable lacuna since it is only in confined systems that the Poincaré recurrence theorem holds and is the main reason why a satisfactory explanation of time's arrow eluded Boltzmann [3] and so many since. We need to see if lifting the phase-space bound gives the sine qua non for solution

[9]: Barbour (2022), 'Time's Arrow and Simultaneity: A Critique of Rovelli's Views'

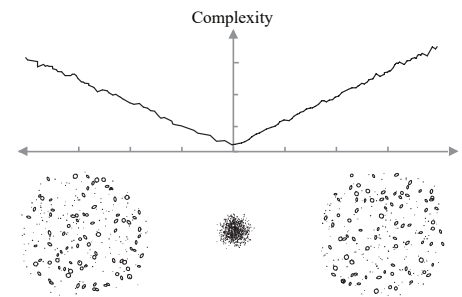
18: It is important to distinguish systems that are closed in the sense of not being subject to external forces from ones that are confined, either physically by a box in a laboratory or theoretically by an artificial nondynamical potential wall. The N-body problem, which I discuss in this paper, is closed in the traditional sense but not confined because critically, as I will emphasise, its scale variable can grow without bound. General relativity, which I also discuss, has spatially closed solutions, but their volume and Hubble radius grow without bound in eternally expanding solutions.

to the problem of time's arrow: time-irreversible behaviour in generic solutions of time-reversal symmetric equations without invocation of a special condition.

He goes on to show this is the case in the oldest dynamical theory, the Newtonian N-body problem of gravitating point particles of masses $m_i, i = 1, \dots, N$ has an unbounded phase space.

In the next section, we take Julian's idea of a Janus point, and instead of treating it as a gravitational arrow of time [ref] , as he does, we instead treat it as a Poincare recurrence *inside* the system of a photon clock. And, perhaps surprisingly, discover that all the same conclusions remain true:

- ▶ "the entire objective content in any N-body solution resides entirely in the shapes it contains." ¹⁹
- ▶ "These 'Kepler pairs' generally live forever and become ever better rods, clocks and compasses all in one, each of them perfectly synchronised with all the others. Unlike growth of entropy, which is associated with increase of disorder, growth of complexity reflects increase of order."
- ▶ "the solutions with negative energy either have a 'Janus region' with bidirectional arrows of time either side of it or remain permanently bound and are virialised within a bounded region as is typical of statistical-mechanical systems with phase spaces of bounded measure. "
- ▶ "This is all a direct consequence of Newton's laws. Despite being time-reversal symmetric, they are fully capable of having solutions that are time-asymmetric on one or both sides of a distinguished point that has a dynamical as opposed to ad hoc origin and show that time-reversal symmetric dynamics is perfectly compatible with a dynamical (not statistical) arrow of time."
- ▶ "There is no distinguished notion of simultaneity in spacetime which can be foliated completely freely by three dimensional hypersurfaces provided they are spacelike.
- ▶ Dirac ... "came to the remarkable conclusion that this could be done "only at the expense of giving up four- dimensional symmetry" and continued "I am inclined to believe from this that four-dimensional symmetry is not a fundamental feature of the physical world." In contrast to Einstein's great insight that "each individual solution ... exhibits four-dimensional symmetry" Dirac argued this need not apply in a quantum theory because "the individual solution has no quantum analogue".
- ▶ " Therefore "Hamiltonian methods ... if expressed in their simplest form, force one to abandon the four-dimensional symmetry"."
- ▶ "Dirac's & York's work ... suggests a need for a qualification of Rovelli's statement "Objectively defined global simultaneity surfaces are not defined in general relativity." ... " since the currently observed accelerated expansion of the universe suggests that, like the N- body problem, it too has an unbounded phase space;"



19: "Imagined as 'snapshots' showing the relative positions of the particles, the shapes could all be jumbled up in a random heap *without any physical information being lost.*"

11 Causal Square-Photon Boxes

Add analysis of Witten's Causal Diamond from other document.

12 Subtime

It is easy to think of the scattering machine has having vastly more surface area to reflect projectiles, and for the infinitesimally small point as having an infinitesimally small probability of occurring in some finite (or infinite) series of experiments. But in a quantum system, the exact opposite could be true.

The photon could get trapped bouncing back and forth 'in perpetuity' between the two points.

If the system measures time by the tick-tock'ing of the photon bounces, then we will find that we have "lost" the ability to measure time (from the outside). Because the events themselves are conserved information within the system.

This is not an enigma. This "hidden variable" behavior is exactly what we see in the laboratory with experiments with photons.

What Beggs & Tucker have missed, is that there is no event that you can use to detect the 'o' condition → by definition, it wasn't captured by either the left detector or the right detector. So there are only two possibilities:

1. The projectile / particle / photon left the apparatus, and went on to contribute its entropy to the universe.
2. The projectile / particle / photon remains trapped within the apparatus, in Subtime.[4, 5, 7].

The subtime essay is now 10 years old, and it's predictions appear consistent with many experiments in the laboratory. Most recently [10] where photons have a tendency to bunch together. Although they now talk about "photonic Temporal Bandwidth".

Add figures – two Wedges back to back with detectors around the side

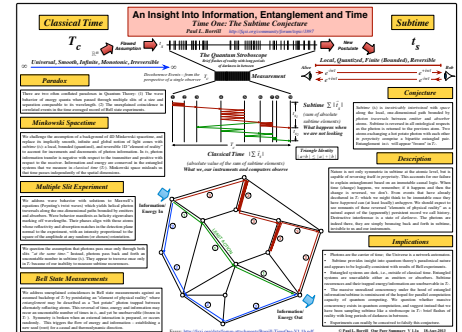
Add Youtube videos on double slit experiment

Veritasium The Original Double Slit Experiment

13 Distributed Autonomous Ethernet

In the DAE Link, we assume two sets of registers in each agent. One register is the 'last' computed value before the message bits go out on the wire, and the other for the 'last' measurement of bits coming in from the wire – for both Alice and Bob. Alice's transmit register is connected by an Ethernet channel to Bob's receive register, and Bob's transmit register is connected by an Ethernet channel to Alice's receive register. This allows us (through one or more sets of transformations – rewriting rules – to create a complete causal loop, with Alice creating events in Bob, and Bob creating Events in Alice²⁰

Our DAE link is a direct consequence of Jonathan' Model. When you connect the circuit of two Jonathan Models, you get A DAE link. When



[10]: Triggiani et al. (2023), 'Ultimate Quantum Sensitivity in the Estimation of the Delay between two Interfering Photons through Frequency-Resolving Sampling'

20: This is not a CTC in GR, this is a Quantum channel with a circulating (alternating) cause and effect [11] that, once kickstarted, will go on forever: "in perpetuity" as we say in the Subtime Conjecture, discussed at the end.

you connect the $\mathbb{N} \rightarrow \mathbb{N}$ and define the arrow, we can use abstract rewriting theory. But unlike the typical way it's done in the Wolfram models, where the theory brings a system back together again with confluence under its own steam, we have an explicit reversal controlled by ancilla.

The DAE Link provides Local Operations and Shared Randomness (LOSR), and described by David Schmid et al. [12, 13]. The Alternating causality protocol allows us to exploit the race conditions inherent in the link as a resource (randomness), by doing this locally (on a shared Ethernet Link) we guarantee that this resource is controllable, and provides – “determinism on demand”.

Each side of the link cannot know what the other side is doing, but by having the Alternating Causality (AC) protocol provide entanglement in the link, we can ‘prime’ the pump of this relationship, such that:

1. If Alice get's there first, she sets the direction of causality.
2. if Bob get's there first, he sets the direction of causality.
3. If Alice and Bob both arrive simultaneously, then either:
 - a) A third party (Carol) gets to arbitrate what she sees in the ordering
 - b) A third party (Captain/Charlie) gets to decide what the ordering should be, according to application constraints, and "turns back time" on one or more of the parties, so that a new (compatible) ordering can take place.

Review: A Functorial Perspective on (MultiComputational) Irreducibility

This section is an addition to the above review. It includes the more expanded article by Jonathan [14].

Here, we look for more potential references to the three possible errors above. The Minkowski Error, the Initial Condition Error, the Reversibility Error, and add a 4th Error : The synchronous CA Error.

We need a better terminology to the above described as errors.

Jonathan is a far better mathematician and theoretical physicist than I am. It's highly likely that I am mistaken in these matters more than Jonathan (Or Edward, or Stephen). Nonetheless, in the spirit of a Work In Progress (WIP) which I plan to use with other theoretical physicists, and computer scientists, here is my analysis, and (I wish I could describe this with more humility, but I don't want to have to expose these ideas to ChatGPT before we expose our IP).

So Here goes ...

THIS IS WHERE I AM UP TO.

TBD - Complete reading of Jonathan's Functorial Perspective paper. Add analysis, references, and cross-check.

[12]: (), *Understanding the interplay of entanglement and nonlocality: motivating and developing a new branch of entanglement theory*

[13]: Schmid et al. (2021), *Unscrambling the omelette of causation and inference*

A Functorial Perspective on (Multi)computational Irreducibility

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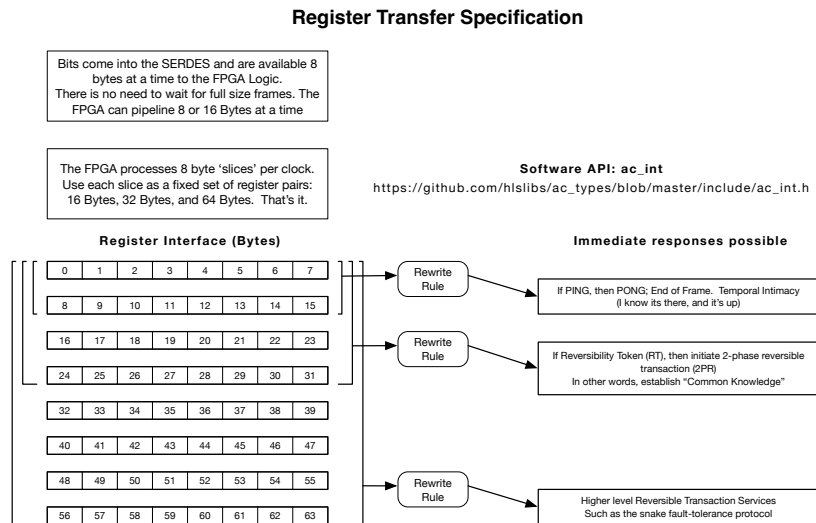
January 13, 2023

Abstract

This article aims to provide a novel formalization of the concept of computational irreducibility in terms of the exactness of functorial correspondence between a category of data structures and elementary computations and a corresponding category of (1-dimensional) cobordisms. We proceed to demonstrate that, by equipping both categories with a symmetric monoidal structure and considering the case of higher-dimensional cobordism categories, we obtain a natural extension of this formalism that serves also to encompass non-deterministic or “multiway” computations, in which one quantifies not only the irreducibility in the behavior of a single (deterministic) computation path, but in the branching and merging behavior of an entire “multiway system” of such paths too. We finally outline how, in the most general case, the resulting symmetric monoidal functor may be considered to be adjoint to the functor characterizing the Atiyah-Segal axiomatization of a functorial quantum field theory. Thus, we conclude by arguing that the irreducibility of (multi)computations may be thought of as being dual to the locality of time evolution in functorial approaches to quantum mechanics and quantum field theory. In the process, we propose an extension of the methods of standard (monoidal) category theory, in which morphisms are effectively equipped with intrinsic computational complexity data, together with an algebra for how those complexities compose (both in sequence and in parallel, subject to the monoidal structure). Some possible extensions of this formalism (for instance to encompass notions of causality, space complexity, irreversibility, complexity of index manipulation in tensor networks, etc.), as well as potential implications for physics (for instance by providing an ability to distinguish formally between certain computational and multicomputational definitions of entropy) are briefly discussed.

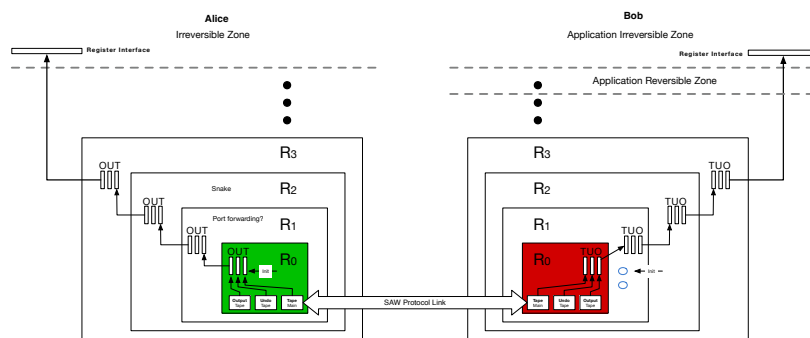
[14]: Gorard (2022), *A Functorial Perspective on (Multi)computational Irreducibility*

14 Register Transfer Logic



This is a specification for our slice engine @ DAE. Those registers (and their slice interactions) can be used in the FPGA to create measurements that are not only orders of magnitude faster than anything that can be achieved in the host processor – with Lingua Franca – but which keep perfect ‘synchrony’ and provide ‘determinism on demand’ with it’s Alternating Causality (AC) mechanism as a resource to manage serializable causality.

Multi-Level, Multi-Way, Reversibility Model



Multi-Layer Logical-Reversibility. Separate layers of Reversible Turing Machines – each with their own Tape, Output, and Undo

15 Forget Time

We’ve known for a long time from Quantum Gravity research by Lee Smolin[15] Carlo Rovelli[16] and others [Cite], that notions of time as a one dimensional line from $-\infty$ to $+\infty$ cannot be correct. This is reflected in our conversation on Friday with Edward referring to his conversation with Eli and reference to the origin of the universe as the boundary condition.

But the real universe is not an empty Minkowski spacetime²¹. We have a different boundary condition in the Symmetric Scattering Machine (photon clock example). The pair of boundary conditions (Reflections of the photon between Alice and Bob).

When there is no "background of time", we need one clock to measure something about another clock (Get Pauli Erker refs).

ADD FIGURE FROM Time Symmetry.

²¹: An empty Minkowski Spacetime would be perfectly smooth and flat

Conversation Pieces

[UNFINISHED - GET MORE PIECES FROM THE TRANSCRIPT]

Transcript

I don't want to interrupt but I'm afraid that I will lose the thought it please. So again, I think But this, this perspective is a little bit too much focused on third party observers and discounts the possibility of interaction. And at a certain level, you're you're absolutely right, that any, any observation of a physical system is going to be limited to a computational realm. And, and this is a, this is a consequence of the Shannon channel capacity theorem, (right?) Because, you know, any communication over a noisy channel conveys a finite number of bits of information. And so, the question then becomes, I actually had kind of an interesting debate with Eli Yovanovitch, who's photonics physicist in my department about this, but, you know, he is a proponent of digital physics, and I was using this is an argument against it. And, and he said, Well, I, he believed that noiseless measurement was, in fact possible. And so I kind of set out to show that, that that's an untestable hypothesis. Okay, and that it's actually it really only follows from circular reasoning that noiseless measurement is only possible if, in fact, digital physics is a valid view of the physical world. And otherwise, it isn't possible. And I constructed a proof of an incompleteness proof for deterministic systems. If you have a rich enough set of deterministic systems, they're topologically. Incomplete, they don't. The set does not contain its own limit points. So from that perspective, if your model is rich enough to describe phenomenon, the physical world in fact what my criteria in there was, it was rich enough to describe Newtonian mechanics, okay, then non determinism is unavoidable. And consequently, noiseless measurement is, is impossible.

Well, it's actually if, but this relates to the the inclusion of continuums, right? So my model is rich and often to describe Newtonian mechanics and the founding mechanics is based on continuums. Right, so the continuums tie into the completeness issue. But they, they also tie into the digital physics issue, right? The the digital physics proponents reject continuums as a valid model of the world. Okay. And so, you know, in a sense, as soon as you do that, my incompleteness result is no longer valid. Okay, because it's not talking about that world. You have a computational model, that's a model of the world that is more limited. And it does not include Newtonian mechanics. Right? As soon as you reject continuums, you don't have Newtonian mechanics anymore. You also don't have relativity. I, on the question of quantum mechanics, I had an interesting discussion with Eli about that, right? He said, he said to me, I, you know, I pointed out the Schrödinger equation relies on continuums. And he said, Yeah, but, you know, if you have any system with well defined boundary conditions, there's only a finite number of solutions to the Schrödinger equation. And I said, Okay, what are the boundary conditions? Right, where, how do you how do you get a physical system with well defined boundary conditions? And he said, well, the universe is finite in extent. Therefore, you have well defined boundary conditions, and there's a finite number of solutions to the Schrödinger equation. Right? So I certainly did his his position on digital physics, on the universe being finite in extent, right. And if it isn't finite in extent, the whole argument breaks down so well, okay.

16 Real Numbers are not Real

Real numbers are not real.

The Borel probability measure is described both by Edward Lee in his Plato and the Nerd, and by Beggs & Tucker in their paper.

Jonathan also uses this in his paper [1].

References to add (From Edward's Ptolemy page).

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Therapy session with ChatGPT

A homomorphism is a structure-preserving map that preserves the operations between two algebraic structures. An isomorphism is a special type of homomorphism that establishes a one-to-one correspondence and preserves all algebraic properties, effectively making the structures indistinguishable from an algebraic perspective.

Lets talk about Category theory and mathematics in general

In mathematics, particularly in algebraic structures such as groups, rings, and vector spaces, homomorphism and isomorphism are important concepts that describe the relationship between two mathematical structures. Let's define each term and explore their differences:

Homomorphism:

A homomorphism is a structure-preserving map or morphism between two algebraic structures. More formally, let $(A, *)$ and $(B, \cdot)$ be two algebraic structures of the same type. A function $f: A \rightarrow B$ is called a homomorphism if it preserves the operations of the structures. That is, for all elements a, b in A , the following holds:

$$f(a * b) = f(a) \cdot f(b)$$

Here, $*$ and $\cdot$ represent the binary operations in structures A and B , respectively. The equality above indicates that the function f preserves the operation of combining elements.

Homomorphisms are useful because they preserve the essential algebraic properties of the structures, such as the group operation or the ring multiplication. They allow us to study the relationship between different structures and transfer properties and concepts from one structure to another. Isomorphism:

An isomorphism is a special type of homomorphism that establishes a one-to-one correspondence between two structures while preserving all algebraic properties. Formally, an isomorphism is a bijective homomorphism. If there exists a homomorphism $f: A \rightarrow B$ that is both injective (one-to-one) and surjective (onto), then we say that A and B are isomorphic, denoted by $A \cong B$.

When two structures are isomorphic, they are essentially the same from an algebraic perspective, despite potentially having different underlying elements. Isomorphic structures share all algebraic properties, such as the group axioms, ring properties, or vector space properties. Consequently, any statement or property that holds for one structure also holds for the isomorphic structure.

In summary, a homomorphism is a structure-preserving map that preserves the operations between two algebraic structures. An isomorphism is a special type of homomorphism that establishes a one-to-one correspondence and preserves all algebraic properties, effectively making the structures indistinguishable from an algebraic perspective.

NOTE:

The notations $\cong$ and $\simeq$ are not totally standardized. Both are usually used for "isomorphic" which means "the same in whatever context we are." For example "geometrically isomorphic" usually means "congruent," "topologically isomorphic" means "homeomorphic," et cetera: it means they're somehow the "same" for the structure you're considering, in some senses they are "equivalent," though not always "equal:" you could have two congruent triangles at different places in a plane, so they wouldn't literally be "the same" but their intrinsic properties are the same. I've seen colleagues use both for isomorphic, and some (mostly the stable homotopy theorists I hang out with) will use $\cong$ for "homeomorphic" and $\simeq$ for "up to homotopy equivalence," but then others will use the same two symbols, for the same purposes, but reversing which gets which symbol. The $\approx$ is used mostly in terms of numerical approximations, meaning that the values in questions are "close" to each other in whatever context one is working, and often it is less precise exactly how "close." Topologists also have a tendency to use $\approx$ for homeomorphic. The main take-away from this answer: notation is not always standardized, and it's important to make sure you understand in whatever context you're working. –Adam Hughes

Subadditivity

Subadditivity is a property that applies to functions or measures defined on a set or space. It describes a specific kind of behavior of functions with respect to addition.

Let's consider a function or measure f defined on a set or space X . The function f is said to be subadditive if, for any two subsets A and B of X , the following inequality holds:

$$f(A \cup B) \leq f(A) + f(B)$$

In other words, the value of the function or measure on the union of two subsets is always less than or equal to the sum of the values of the function on the individual subsets.

Intuitively, subadditivity captures the idea that the combined value of the function on two subsets is never greater than the sum of their individual values. It implies that the function exhibits a form of "weak" or "partial" additivity.

Subadditivity is a desirable property in various contexts. In mathematics, it often appears in the study of measures and set functions. For example, in measure theory, a measure that satisfies the subadditivity property is called a subadditive measure. Subadditive measures are commonly used in the study of probability theory, where they provide useful bounds and inequalities for probabilities.

In economics and optimization theory, subadditivity is also relevant. It is often associated with cost functions or utility functions that exhibit diminishing marginal returns. Subadditivity implies that the cost or utility associated with a combined set of goods or activities is less than or equal to the sum of their individual costs or utilities.

Overall, subadditivity is a concept that captures the idea of partial additivity and finds applications in various mathematical and applied fields.

Monoidal Functoriality

Monoidal functoriality refers to the property of a functor between monoidal categories. To understand this concept, let's first define the key components involved:

Monoidal Category: A monoidal category is a category equipped with a tensor product operation that combines objects and a tensor unit object that serves as an identity for the tensor product. This tensor product operation should be associative and have certain coherence properties.

Functor: A functor is a mapping between categories that preserves the structure and relationships between objects and morphisms. It associates objects and morphisms from one category (the source category) to another category (the target category) while respecting the composition of morphisms and preserving identities.

Now, let's delve into monoidal functoriality. A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ between monoidal categories $\mathcal{C}$ and $\mathcal{D}$ is said to be a monoidal functor if it preserves the monoidal structure. This means that the functor F should not only preserve the objects and morphisms of the categories but also respect the tensor product and the tensor unit.

More specifically, for a monoidal functor $F : \mathcal{C} \rightarrow \mathcal{D}$ to be considered monoidal, it needs to satisfy the following conditions:

F preserves the tensor product: For any objects X and Y in $\mathcal{C}$, the functor F should assign to their tensor product $(X \otimes Y)$ in $\mathcal{C}$ the tensor product $(F(X) \otimes F(Y))$ in $\mathcal{D}$.

F preserves the tensor unit: The functor F should map the tensor unit object $(I_{\mathcal{C}})$ in $\mathcal{C}$ to the tensor unit object $(I_{\mathcal{D}})$ in $\mathcal{D}$.

F preserves the tensor product associativity and coherence: The functor F should respect the associativity isomorphisms and coherence isomorphisms in both the source and target categories.

By preserving the monoidal structure, a monoidal functor enables the translation of algebraic operations and relationships between objects in the source category to corresponding operations and relationships in the target category, while maintaining consistency with the tensor product and tensor unit. This property is useful for studying and analyzing monoidal categories and their relationships through the lens of functorial mappings.

Braiding

Braiding refers to a concept that arises in the theory of braided categories and braided monoidal categories. A braided category is a category equipped with a braiding operation that allows for the interchanging of objects in a specific way.

More specifically, given a braided category, the braiding is a natural isomorphism that operates on pairs of objects within the category. It provides a way to exchange the order of objects while respecting the categorical structure.

The braiding is typically denoted by the symbol σ or c , and it satisfies certain properties. For a pair of objects A and B in a braided category, the braiding σ operates as follows:

$$\sigma : A \otimes B \rightarrow B \otimes A$$

Here, $\otimes$ represents the tensor product operation within the category. The braiding σ allows for the interchange of objects A and B by transforming their tensor product $A \otimes B$ into $B \otimes A$. The precise properties that the braiding must satisfy are known as the braid equations or the Yang-Baxter equations, which express the consistency and compatibility of the braiding operation.

Braided categories and the associated braiding operation have applications in various branches of mathematics, including knot theory, topological quantum field theory, and quantum computation. The braiding plays a crucial role in capturing and studying the non-commutative nature of these mathematical structures and providing a framework for understanding the behavior of intertwined objects.

Cobordism

Cobordism is a concept that originated in algebraic topology, specifically in the study of manifolds. Cobordism provides a way to classify and study the relationship between manifolds in terms of their boundaries.

A cobordism between two manifolds is a higher-dimensional manifold whose boundary is precisely the disjoint union of the given manifolds. In other words, it serves as a "bridge" connecting the two manifolds. The term "cobordism" comes from the idea of a "coboundary," which is a boundary of a boundary.

More formally, let's consider two closed (compact without boundary) manifolds M and N of the same dimension. A cobordism between M and N is a compact manifold W , often with additional structure, such as a smooth or topological structure, such that the boundary of W is precisely the disjoint union of M and N .

The study of cobordism provides a powerful tool for understanding the topology of manifolds. Cobordism theory involves classifying cobordism classes, which are equivalence classes of cobordisms between manifolds. Two cobordisms are considered equivalent if they can be smoothly deformed into each other while keeping the boundary fixed.

Cobordism theory has important applications in various areas of mathematics, including algebraic topology, geometric topology, and geometric analysis. It provides a framework for understanding the relationships between manifolds and has connections to other topics such as characteristic classes, surgery theory, and the study of topological invariants.

Strict Positivity

In mathematics, strict positivity refers to a property or condition that states that a certain quantity or object is strictly greater than zero, without the possibility of being equal to zero. It implies that the quantity or object must be positive and not approach or reach zero in any way.

The concept of strict positivity is often encountered in various mathematical contexts, including real analysis, linear algebra, optimization, and probability theory. It is particularly relevant when dealing with inequalities, where strict positivity indicates a strict inequality.

For example, consider a real-valued function $f(x)$. If we say that $f(x)$ is strictly positive, it means that $f(x) > 0$ for all x in the given domain, and there are no points at which $f(x)$ equals zero. In this case, the function does not "touch" or approach zero but stays strictly above it.

Similarly, in linear algebra, a matrix or vector is said to be strictly positive if all its entries are strictly greater than zero. This property is important in the study of positive definite matrices and positive cone in convex analysis.

The notion of strict positivity allows for sharper statements and analysis in mathematical theories, as it excludes the possibility of equality to zero, providing a stricter condition for positivity.